Examples for Hypothesis Testing

Example 1: Coral Gable House Sales: Test for Single Popultation Mean

The data being used is the Coral Gables house sales data as being discussed in class. We first read the data in and obtain some key statistics.

```
houses=read.csv("CoralGablesHouses.csv")
n=length(houses$SalePrice)
xbar=mean(houses$SalePrice)
s=sd(houses$SalePrice)
xbar
```

```
## [1] 564973.3
```

Let μ denote the mean sale prices of all houses (suppose that all houses could have been put on the market for sale). We would like to perform three different tests, all with a level of significance equal to 0.01. The hypotheses for the first test are:

$$H_0 : \mu = 500,000, \quad H_a : \mu \neq 500,000.$$

This can be done by using the `t.test()` function.

```
mu0=500000
t.test(houses$SalePrice,alternative="two.sided",mu=mu0)
```

```
##
##  One Sample t-test
##
## data:  houses$SalePrice
## t = 4.1629, df = 535, p-value = 3.662e-05
## alternative hypothesis: true mean is not equal to 5e+05
## 95 percent confidence interval:
##  534313.4 595633.2
## sample estimates:
## mean of x
##  564973.3
```

We write `alternative="two.sided"` because $H_a : \mu \neq 500,000$ is a two-tailed test. The argument `mu=mu0` defines the hypothesized value (=500,000). The p-value is 3.662e-05 which is smaller than 0.01. We therefore reject the null hypothesis.

The hypotheses for the second test are:

$$H_0 : \mu \leq 500,000, \quad H_a : \mu > 500,000,$$

```
t.test(houses$SalePrice,alternative="greater",mu=mu0)
```

```
##
##  One Sample t-test
##
## data:  houses$SalePrice
## t = 4.1629, df = 535, p-value = 1.831e-05
## alternative hypothesis: true mean is greater than 5e+05
## 95 percent confidence interval:
##  539256.4      Inf
## sample estimates:
## mean of x
##  564973.3
```

We write `alternative="greater"` because $H_a : \mu > 500,000$ is a right-tailed test and the `>` sign is used in H_a. The p-value is 1.831e-05 which is smaller than 0.01. We therefore reject the null hypothesis.

The hypotheses for the third test are:

$$H_0 : \mu \geq 500,000, \quad H_a : \mu < 500,000,$$

This can be done again by using the `t.test()` function.

```
t.test(houses$SalePrice,alternative="less",mu=mu0)
```

```
##
##  One Sample t-test
##
## data:  houses$SalePrice
## t = 4.1629, df = 535, p-value = 1
## alternative hypothesis: true mean is less than 5e+05
## 95 percent confidence interval:
##      -Inf 590690.2
## sample estimates:
## mean of x
##  564973.3
```

We write `alternative="less"` because $H_a : \mu < 500,000$ is a left-tailed test and the `<` sign is used in H_a. The p-value is 1 which is larger than 0.01. We therefore fail to reject the null hypothesis.

Exercise

Test whether the mean age of the properties is equal to 50 years old or not.

Example 2: Coral Gable House Sales: Test for Two Popultation Means

The `t.test()` function can be used to obtain confidence intervals and to conduct hypothesis testing for two variables. For example, suppose that we are interested in comparing the mean sale prices for houses located to the east and west of US1.

We first create two new variables for the sale prices for these two different types of properties. We first use the `which()` function to identify the positions of these two types of properties.

```
positions_east=which(houses$eastUS1=="yes")
positions_west=which(houses$eastUS1=="no")
SalePrice_east=houses$SalePrice[positions_east]
SalePrice_west=houses$SalePrice[positions_west]
```

Let μ_e and μ_w denote the mean sale prices for houses located to the east and west of US1, respectively. Suppose we would like to test the following hypotheses.

$$H_0 : \mu_e = \mu_w, \quad H_a : \mu_e \neq \mu_w.$$

Or equivalently,

$$H_0 : \mu_e - \mu_w = 0, \quad H_a : \mu_e - \mu_w \neq 0.$$

To do so, we first test whether the two variances are equal. Let σ_e^2 and σ_w^2 denote the variances for houses located to the east and west of US1. Suppose we would like to test the following hypotheses.

$$H_0 : \sigma_e^2 = \sigma_w^2, \quad H_a : \sigma_e^2 \neq \sigma_w^2.$$

We can use the `var.test()` function to perform the F-test.

```
var.test(SalePrice_east,SalePrice_west,alternative = "two.sided")
```

```
##
##  F test to compare two variances
##
## data:  SalePrice_east and SalePrice_west
## F = 5.1826, num df = 161, denom df = 373, p-value < 2.2e-16
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  4.015225 6.782362
## sample estimates:
## ratio of variances
##           5.182594
```

The above output shows that the p-value is less than $2.2 * 10^{-16}$ which is less than 0.01. Therefore, we reject the null hypothesis that $\sigma_e^2 = \sigma_w^2$.

We now test the two means using the `t.test()` function.

```
t.test(SalePrice_east,SalePrice_west,alternative = "two.sided")
```

```
##
##  Welch Two Sample t-test
##
## data:  SalePrice_east and SalePrice_west
```

```
## t = 7.1272, df = 188.47, p-value = 2.117e-11
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  213881.1 377585.2
## sample estimates:
## mean of x mean of y
##  771324.4  475591.3
```

The above output shows that the p-value is 2.117e-11 which is less than 0.01. We therefore reject the null hypothesis. Moreover, it shows a 95% confidence interval for the mean of the first variable (`SalePrice_east`) minus the mean of the second variable (`SalePrice_west`), i.e., $\mu_e - \mu_w$. Thus, we are 95% confident that properties located to the east of US1 are on average more expensive than those located to the west of US1 by an amount between $213,881.1 and $377,585.2.

More on T-test in R:

- If we conclude that the variances are the same for the two samples, the argument `var.equal=TRUE` can be used to indicate to the `t.test()` function that the two samples have equal variances. The t-test can be performed as `t.test(x,y,var.equal=TRUE)` .

- If we want to perform paired t-test, the argument `paired=TRUE` can be used. For example, `t.test(x,y,paired=TRUE,alternative="two.sided")` .

Example 3: Coral Gable House Sales: Test for Single Popultation Proportion

Let p denote the true proportion of all properties that had a pool. We would like to test whether over 25% of properties had a pool with a level of significance equal to 0.01. The hypotheses are

$$H_0 : p \leq 0.25, \quad H_a : p > 0.25.$$

This can be done by using the `prop.test()` function. We use the `which()` function to find the positions of properties with pools and then use the `length()` function to calculate the number of such properties.

```
p0=0.25
positions_pool=which(houses$Pool=="yes")
x=length(positions_pool)
```

Now we run the `prop.test()` function. We will need to put in the number of yeses (`x`), the total number of properties (`n`), the hypothesized value (`p0`), as well as the alternative (`greater` because the `>` sign in used in $H_a : p > 0.25$).

```
prop.test(x,n,p0,alternative='greater')
```

```
##
##  1-sample proportions test with continuity correction
##
## data:  x out of n, null probability p0
## X-squared = 22.45, df = 1, p-value = 1.078e-06
## alternative hypothesis: true p is greater than 0.25
## 95 percent confidence interval:
##  0.3058824 1.0000000
## sample estimates:
##         p
## 0.3395522
```

The p-value is 1.078e-6 which is smaller than 0.01. We therefore reject the null hypothesis.

Exercise

Test the following hypotheses

$$H_0 : p \geq 0.2, \quad H_a : p < 0.2.$$

where p now stands for the proportion of properties using the city sewer system (`CitySewer==yes`).

Example 4: Coral Gable House Sales: Test for Two Popultation Proportions

The `prop.test()` function can also handle two variables. For example, suppose that we are interested in testing whether the true proportion of properties located to the east of US1 with pool was higher than those located to the west of US1. Let p_{east} and p_{west} denote the two true proportions. The hypotheses are

$$H_0 : p_{east} \leq p_{west}, \quad H_a : p_{east} > p_{west}.$$

Or equivalently,

$$H_0 : p_{east} - p_{west} \leq 0, \quad H_a : p_{east} - p_{west} > 0.$$

We first create two new variables for the sale prices for these two different types of properties. We first use the `which()` function to identify the positions of the different types of properties.

```
positions_east=which(houses$eastUS1=="yes")
positions_east_pool=which(houses$eastUS1=="yes"&houses$Pool=="yes")
positions_west=which(houses$eastUS1=="no")
positions_west_pool=which(houses$eastUS1=="no"&houses$Pool=="yes")

x_east=length(positions_east_pool)
n_east=length(positions_east)
x_west=length(positions_west_pool)
n_west=length(positions_west)
```

Now we apply the `prop.test()` function to the corresponding proportions. We use the function `c()` to combine the counts.

```
prop.test(c(x_east,x_west),c(n_east,n_west),alternative='greater')
```

```
##
##  2-sample test for equality of proportions with continuity correction
##
## data:  c(x_east, x_west) out of c(n_east, n_west)
## X-squared = 78.092, df = 1, p-value < 2.2e-16
## alternative hypothesis: greater
## 95 percent confidence interval:
##  0.3216105 1.0000000
## sample estimates:
##    prop 1    prop 2
## 0.6172840 0.2192513
```

The above output says that the p-value is less than 2.2e-16 which is less than 0.01. We therefore reject the null hypothesis.